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Terminal Illness Claims

Health Claims Forum – 16 May 2013

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Agenda



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- Terminal Illness Claims Study – Introduction
- Study Results – Brendon Gerding, Senior Actuarial Analyst, PLRe
- Practical Claim Management – Paul Reddick, Head of Medical Underwriting & Claims, PLRe
 - Medical Factors
 - Evidence gathering
 - PLRe Philosophy
 - Conclusions
- Cancer, Palliative Treatment and Survival Rates - Dr James D. Brenton
PhD FRCP

The Study – Summary

- We analysed around 2,700 terminal illness claims that were paid under life insurance policies over a period of several years.
- The objective was to determine whether the claimant was alive or dead
- The data is from several companies so reflects a range of claim philosophies

Why undertake a study?

- A unique opportunity to evaluate the accuracy and impact of Terminal illness claim decisions - over a long time period
- To identify and evaluate any patterns or trends
- To see if there are any circumstances where faster claim payments can be made (e.g. Specific cancer sites)



- Terminal Illness claims as a proportion of all death claims have increased over the last 5 years - now account for 25% of all claims
- Payments are based on the likelihood of death occurring within a relatively short time-frame - we want to understand if we are getting it right
- This is a highly sensitive area - could we identify any opportunities to remove obstacles at this difficult time by making faster claims decisions - without compromising accuracy?
- Would there be any cancer diagnoses or other conditions where survival was materially longer than might have been expected?



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Brendon Gerding
Senior Actuarial Analyst

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Introduction

- Aim is to analyse the mortality of the terminally ill (within insurance definition)
- First study of its kind
- Existing hospital studies – not insured population, no “12 month” requirement
- We will look to answer the following:
 - Why perform an analysis?
 - How to analyse?
 - What are the results?

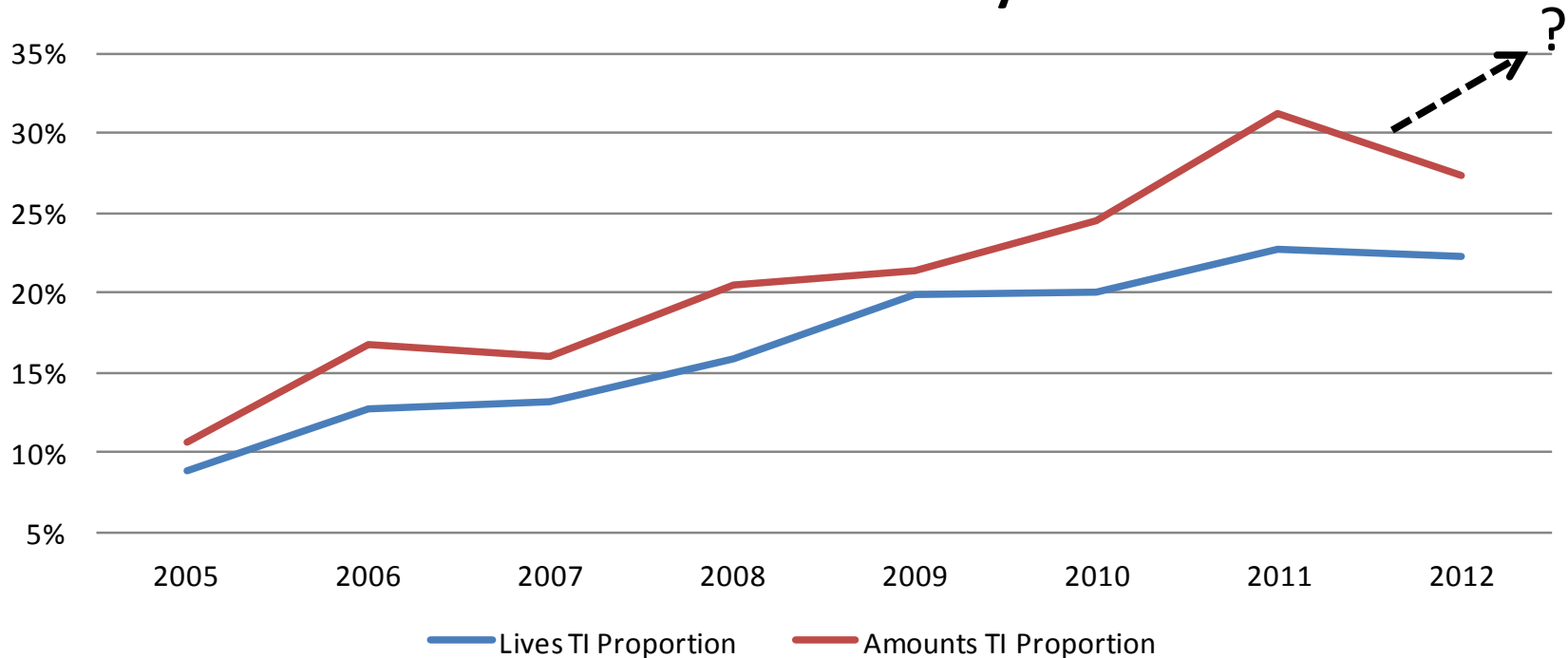
Why analyse?

Terminal Illness Trends



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TI % of Term Assurance Claims by Year of Claim



- Increased awareness over time and for higher sum assureds
- TI declined claims reducing quicker than death claims
- What is the upper limit?

Why analyse?

The cost of Terminal Illness



What is the Cost of Terminal Illness Benefit?

Assume:

- 25% of claims are due to Terminal Illness
- Average TI claimant survives 12 months
- Base group is Males, Non-smokers, Age 40

Approximate cost for having TI included in policy:

Policy Term	Policy Type	
	Level	Decreasing
20 Year	1.4%	2.4%
10 Year	1.9%	5.2%
5 Year	3.3%	12.0%

How to Analyse?

Data preparation



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- TI Claims data sourced from various companies
- Claims are de-duplicated to prevent double counting
- Resultant number of claims:

Year Claim Paid	# TI Claims Paid
2003	7
2004	38
2005	66
2006	70
2007	139
2008	277
2009	507
2010	681
2011	750
2012	116
Total	2,651

How to Analyse?

Claims by cause



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Delay between diagnosis and paying a claim (months):

Cause of Claim	# Claims	% Claims	Avg Delay
Lung	557	21%	2.6
Digestive	304	11%	2.4
Other Cancer	279	11%	2.7
Brain	265	10%	3.0
Bowel	254	10%	2.7
Breast	202	8%	2.3
Urological	199	8%	2.3
Pancreatic	168	6%	2.7
Gynaecological	84	3%	2.1
Motor Neurone Disease	58	2%	1.8
Skin	68	3%	3.1
Haematological	85	3%	2.4
Liver	74	3%	3.2
Other Disease (non-Cancer)	46	2%	2.8
No Data	8	0%	1.4
Totals	2,651	100%	2.6

- Brain, skin and liver cancer claims have high delays
- Breast, urological, gynaecological cancer and MND claims paid quickest

How to Analyse?

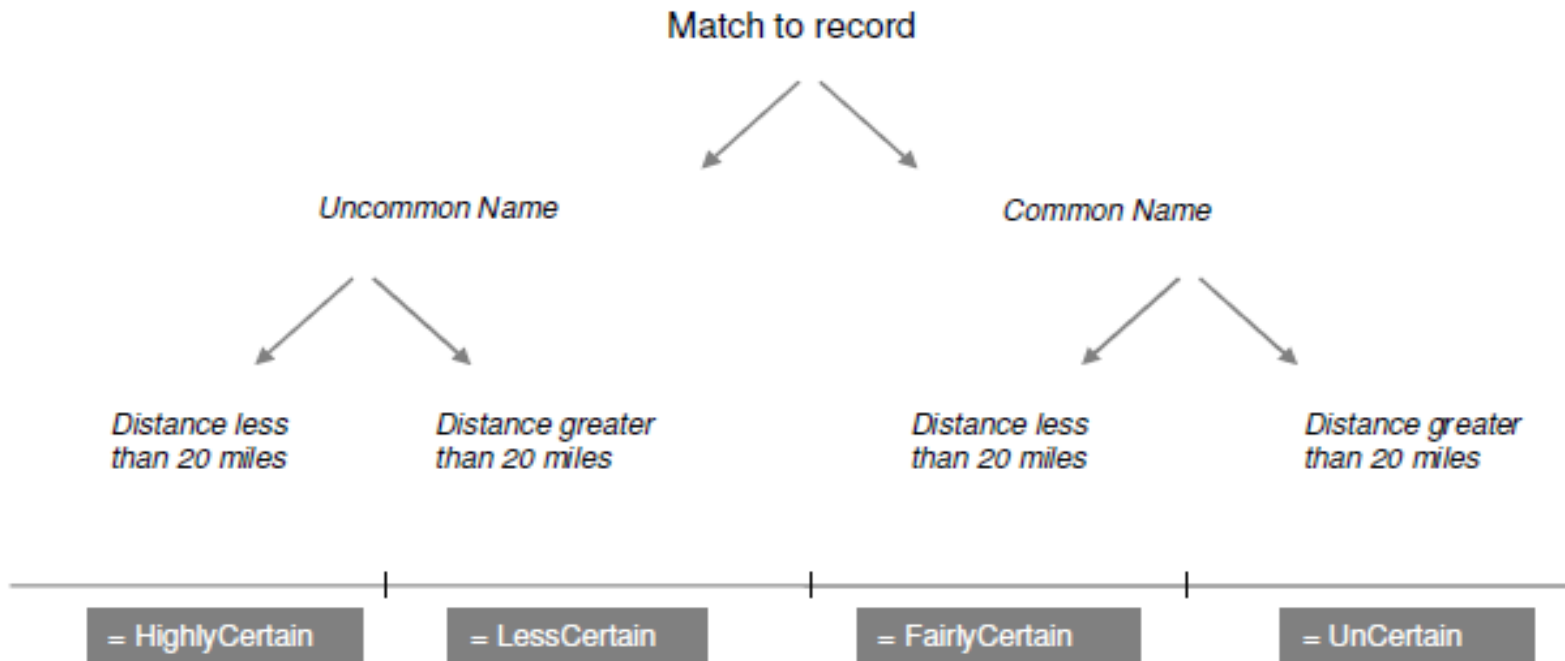
Tracing TI Claimants



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January 2013 – claimants are traced using Capita Tracing

- 1) Requires match to death records by **Surname, Forename, DoB, Gender**
- 2) Then assigns “grading” to the match as follows:



How to Analyse?

Tracing TI Claimants



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Results of Tracing Exercise:

Trace Result	# Claims	% Claims
Highly Certain Death	2,111	80%
Fairly Certain Death	32	1%
Less Certain Death	39	1%
Uncertain Death	87	3%
No Match (Still Alive)	382	14%
Total	2,651	100%

These results don't tell us much about **when** someone is expected to die, or how long those **still alive** have been alive for...

How to Analyse?

Survivorship over Time

Aim

- To incorporate **death and survivorship** statistics into **one single measure**

Methodology

- Examine the **number of people exposed to the risk of dying** at each day following their diagnosis date
- Calculate a **daily death rate**, bearing in mind survivors to end of period (i.e. January 2013)
- Daily mortality rate used to **generate a survivorship curve** over time

Things to bear in mind...

- Aggregates data from all available claim years and companies, therefore **does not allow for changing trends over time**
- “Uncertain” deaths have been excluded – “highly certain”, “less certain” and “fairly certain” are included

How to Analyse?

Survivorship over Time

Methodology example:

Time since dx	# People	# Deaths	# Alive Reaching Study End	Death Rate	Survivorship
0	100	1	0	1.0%	100.0%
1	99	3	2	3.0%	99.0%
2	94	2	4	2.1%	96.0%
3	88	4	5	4.5%	94.0%
4	79	1	2	1.3%	89.7%
5	76	0	3	0.0%	88.6%
...	...	...	...	...	...

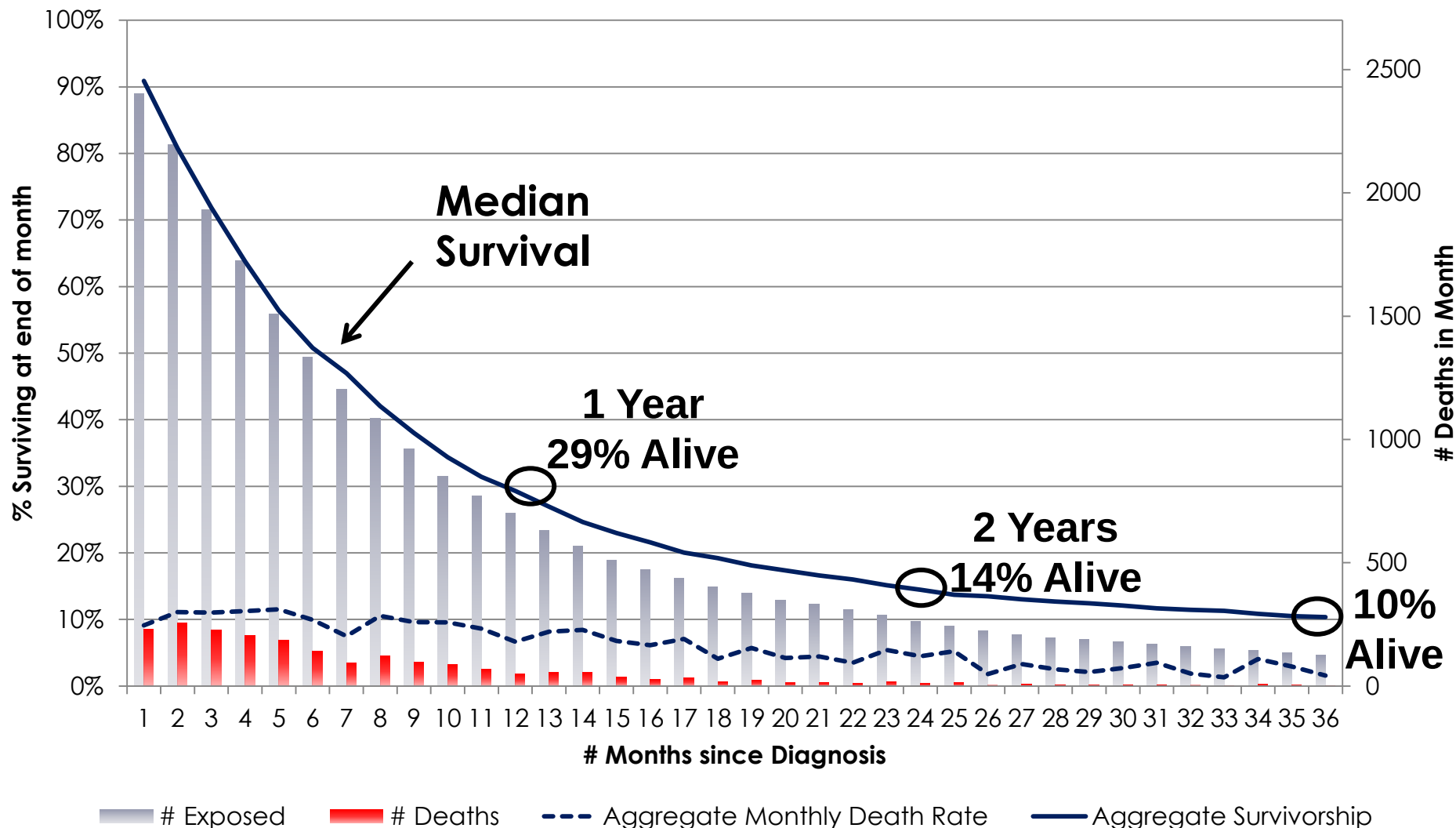
Results

Survivorship over time



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Monthly Survivorship - All Conditions



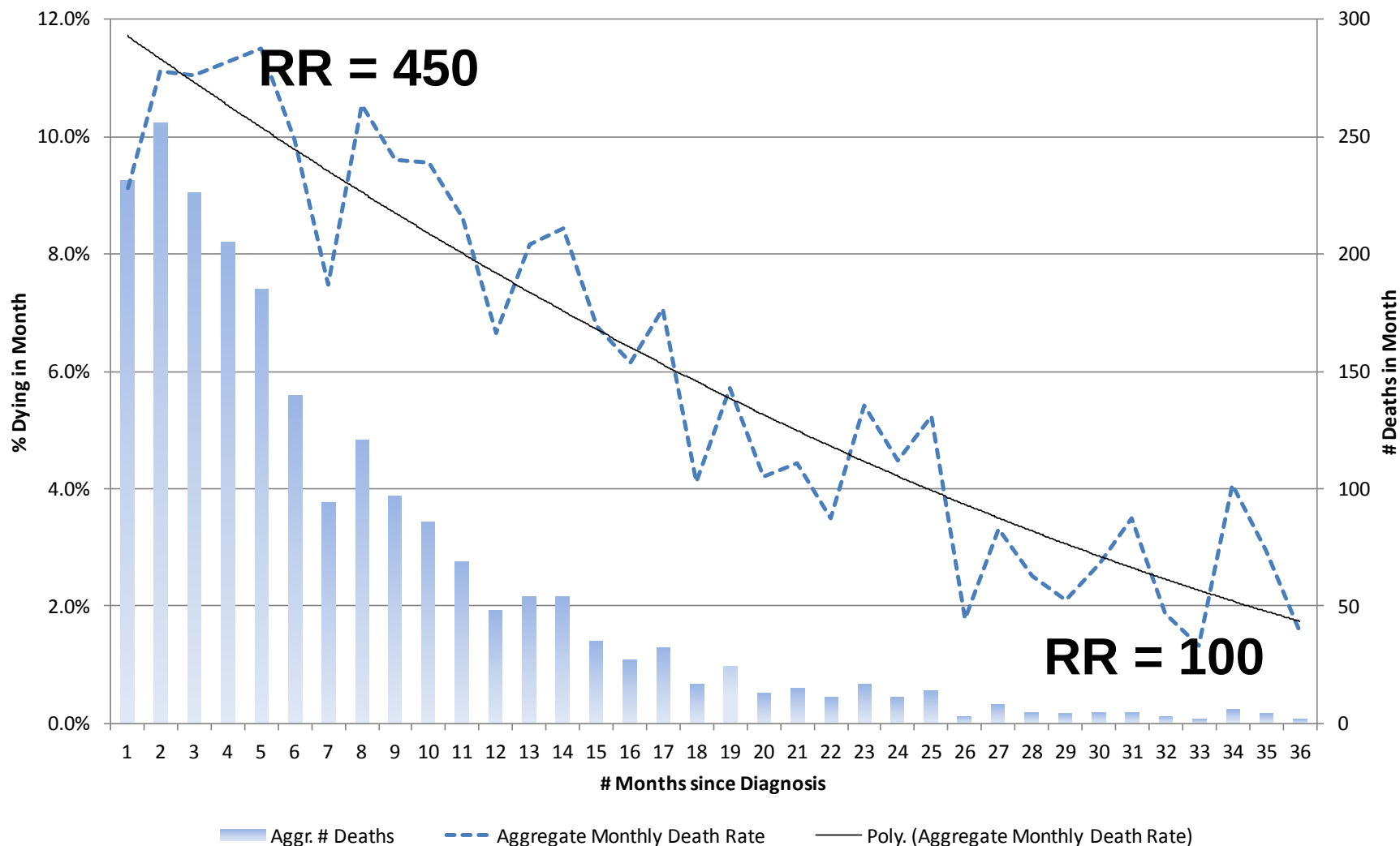
Results

Survivorship over time



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Monthly Death Rate - All Conditions



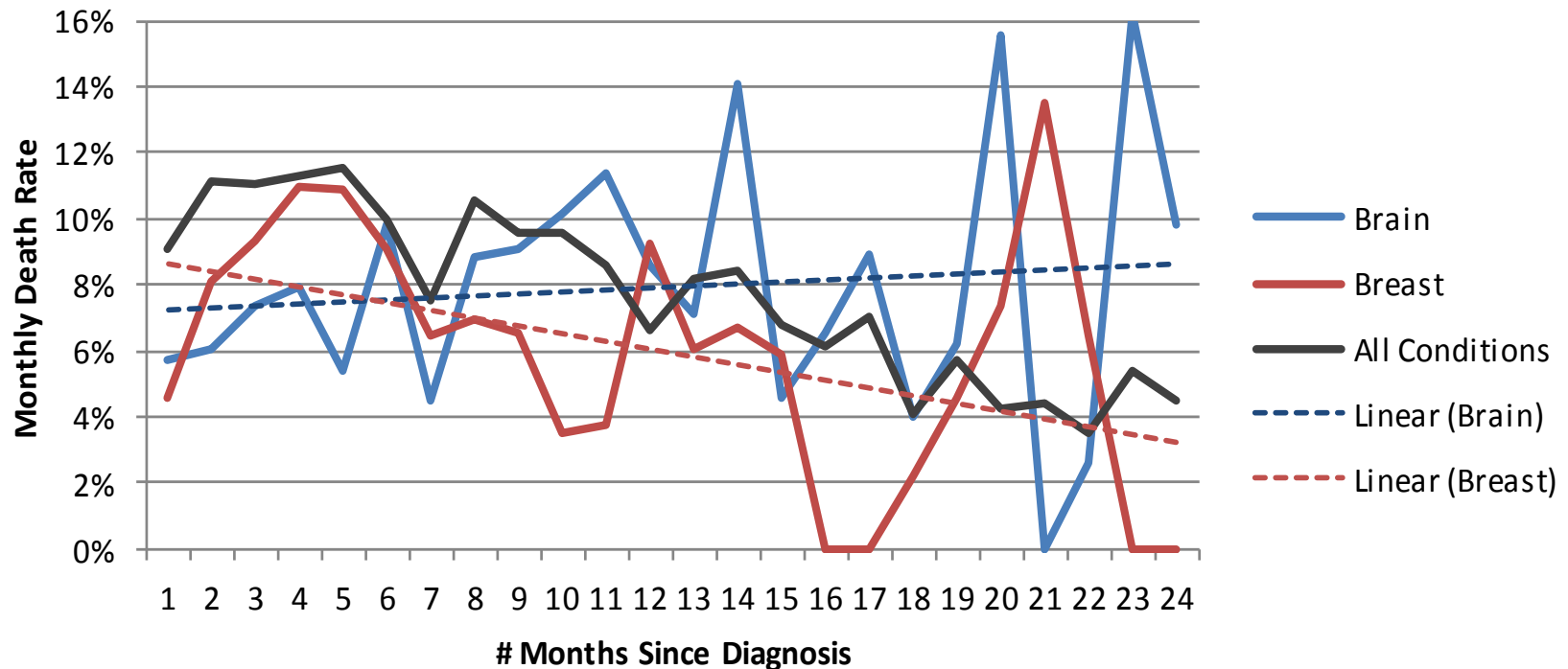
Results

Survivorship over time



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Monthly Death Rate Following Diagnosis



- Apparent difference in progression of mortality following diagnosis
- Interestingly, despite high survivorship, breast cancer has the highest relative risk of all conditions (lower average age, almost all females)

Results

Survivorship over time



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We can summarise the survival statistics:

Survival Measure	Survival Period
Median survival	184 days
Life expectancy of those who died:	284 days



BUT, what about those still alive?

50% of individuals dead. Ignores those still alive after this point (i.e. the “tail”)

Accounts for the length of survival, as well as % surviving

**Example – UK Salaries:
Median salary: £21k
Mean salary: £29k
(Skew towards higher salaries)**

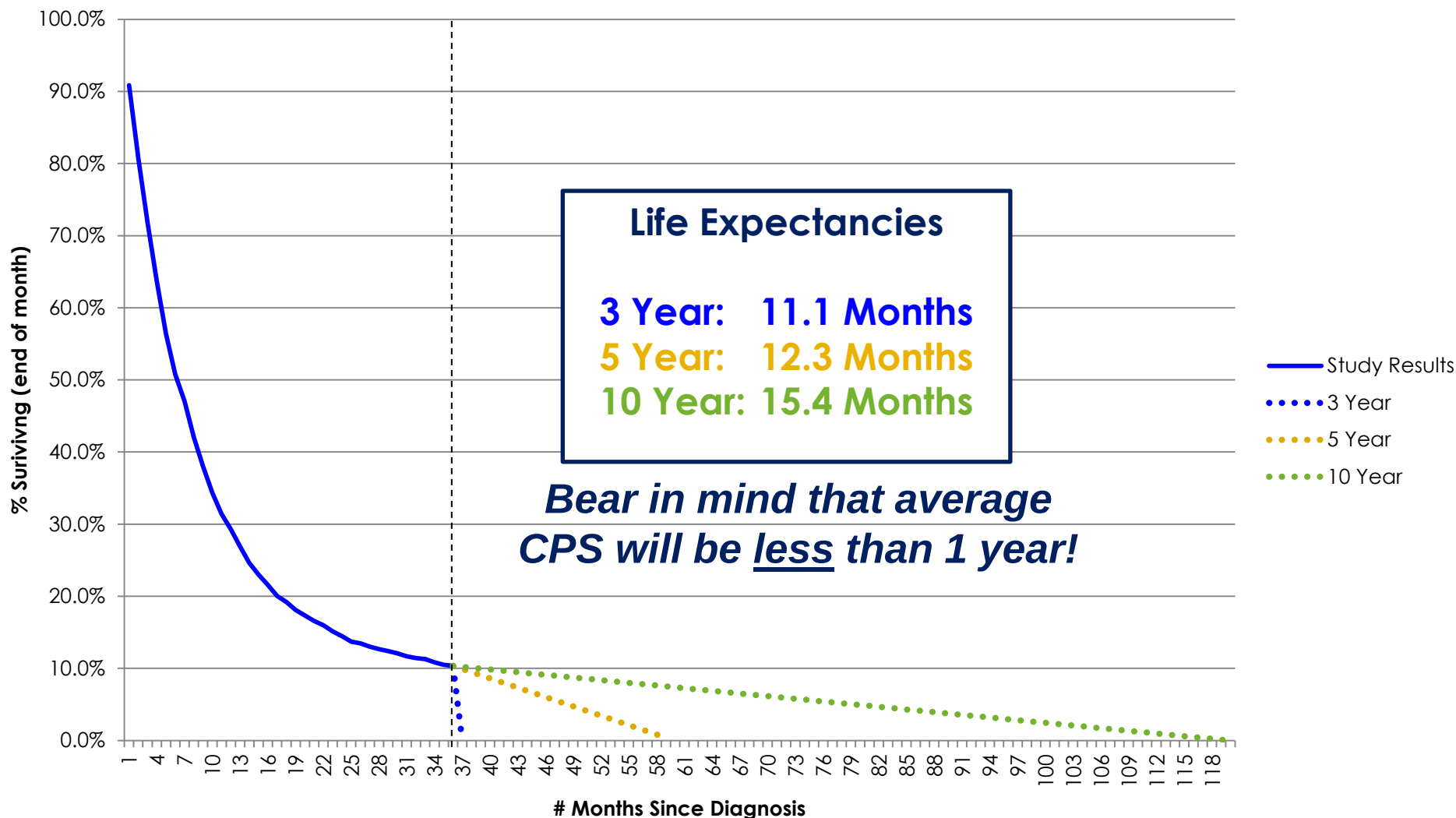
Results

Survivorship over time



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Future Survival Possibilities



Results

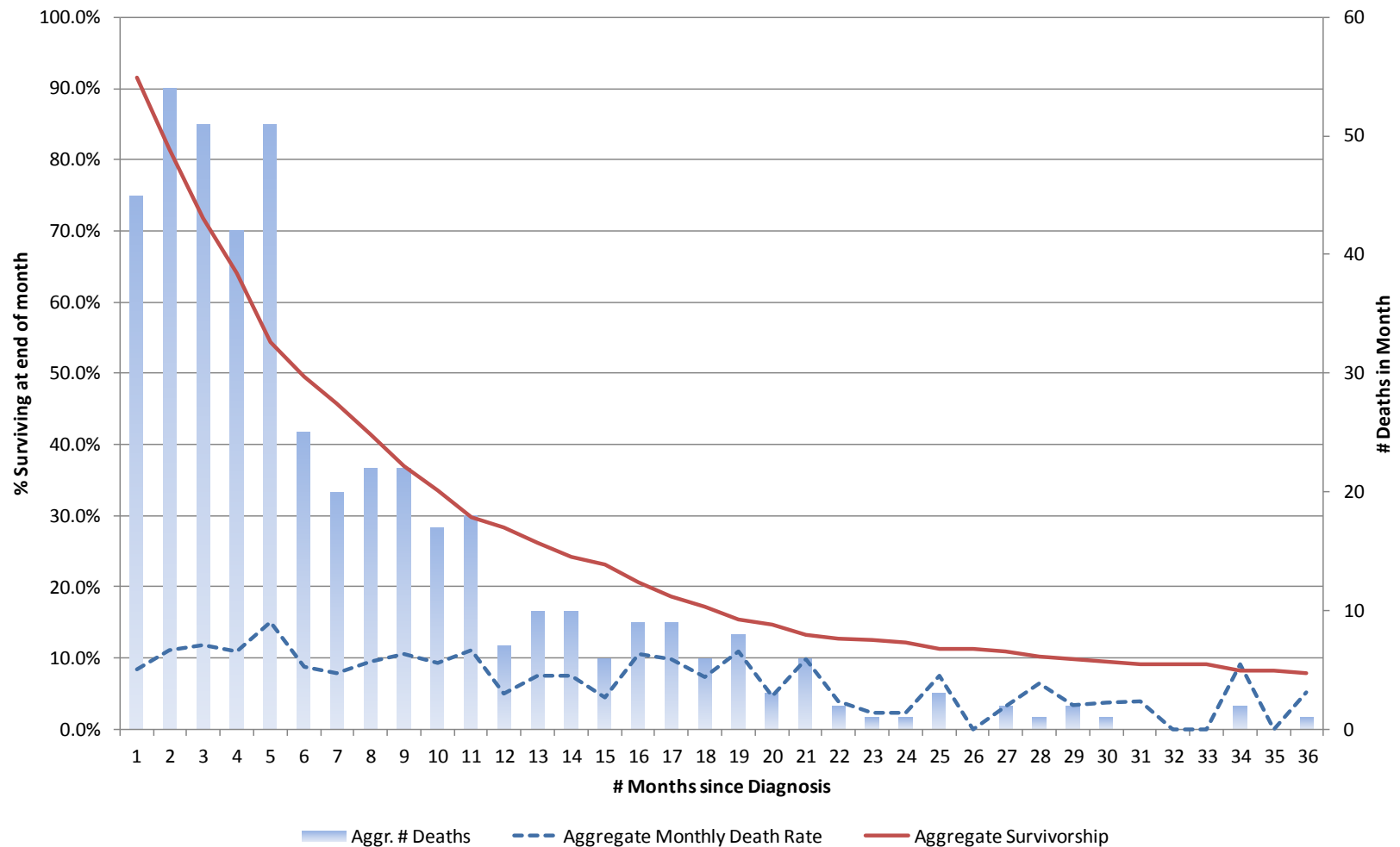
Lung Cancer



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■ 557 Claims, 21% of total

Monthly Survivorship - Lung Cancer



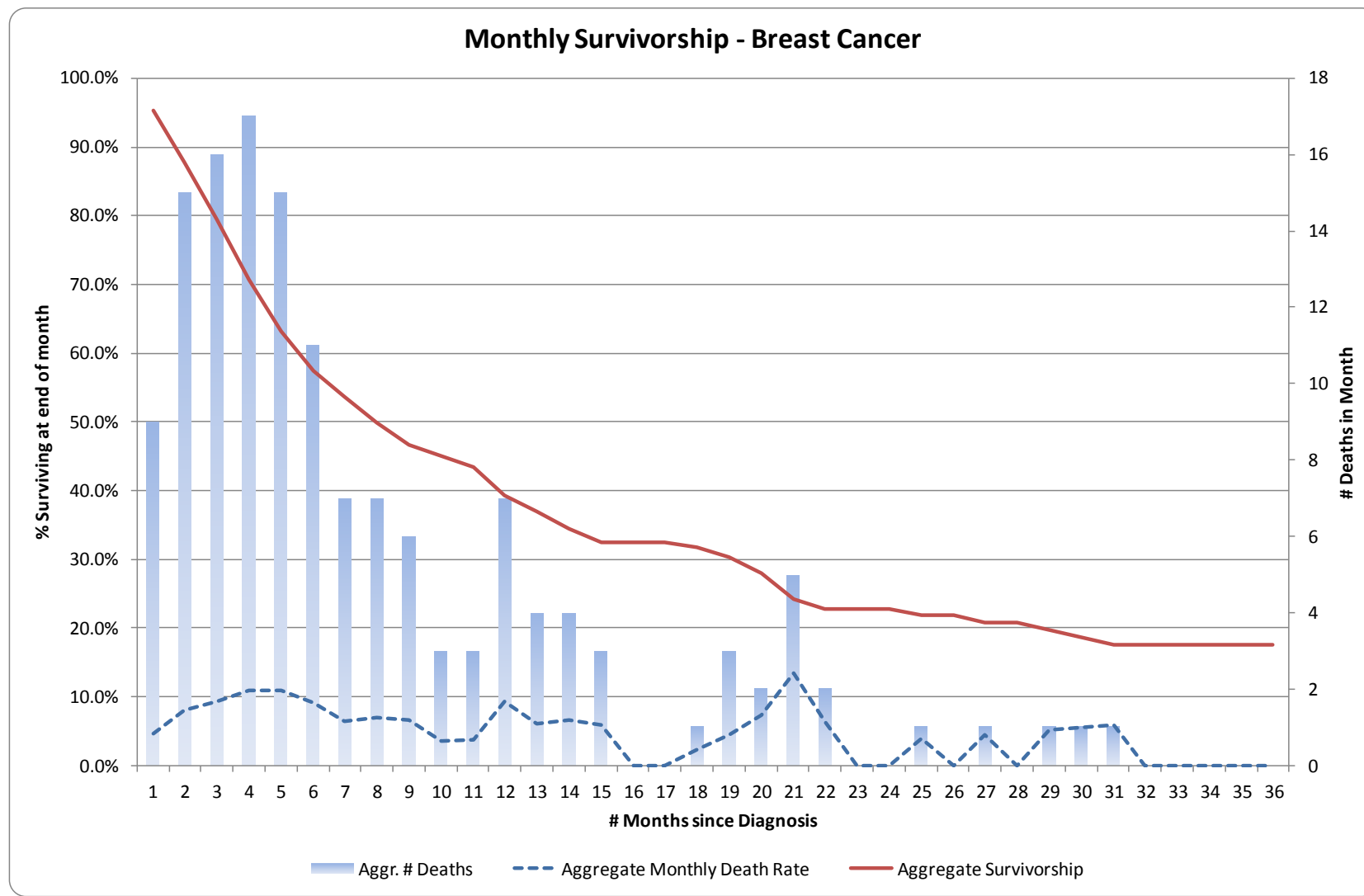
Results

Breast Cancer



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■ 202 Claims, 8% of total



Results

Condition specific survival



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Survivorship results by condition

- “Other cancer” includes gynaecological, liver and skin cancer

Number of Deaths											
Duration since Dx	Lung	Digestive	Brain	Bowel	Urological	Breast	Pancreatic	Haem.	Other Cancer	Other Diseases	All Conditions
Within 12 Months	374	225	151	163	135	116	124	56	358	49	1,758
12-24 Months	71	31	51	34	16	24	13	5	49	13	310
Aggregate	445	256	202	197	151	140	137	61	407	62	2,068
Number Still in Study											
Duration since Dx	Lung	Digestive	Brain	Bowel	Urological	Breast	Pancreatic	Haem.	Other Cancer	Other Diseases	All Conditions
At start	557	302	265	254	199	202	168	80	487	101	2533
After 12 Months	140	55	88	71	47	72	38	19	121	44	699
After 24 Months	42	15	29	20	28	27	14	11	54	24	263
Survival Rates											
Duration since Dx	Lung	Digestive	Brain	Bowel	Urological	Breast	Pancreatic	Haem.	Other Cancer	Other Diseases	All Conditions
12 Months	28%	20%	37%	31%	28%	39%	23%	28%	26%	50%	29%
24 Months	12%	8%	13%	13%	18%	23%	13%	20%	14%	33%	14%

- Data is sparse for durations after 2 years
- Highest survivorship (1yr): Brain, Breast, Non cancers
- Highest survivorship (2yr): Urological, Breast, Haematological, Non cancers
- Lowest survivorship (1yr): Digestive, Pancreatic
- Lowest survivorship (2yr): Digestive, Lung, Brain, Bowel, Pancreatic

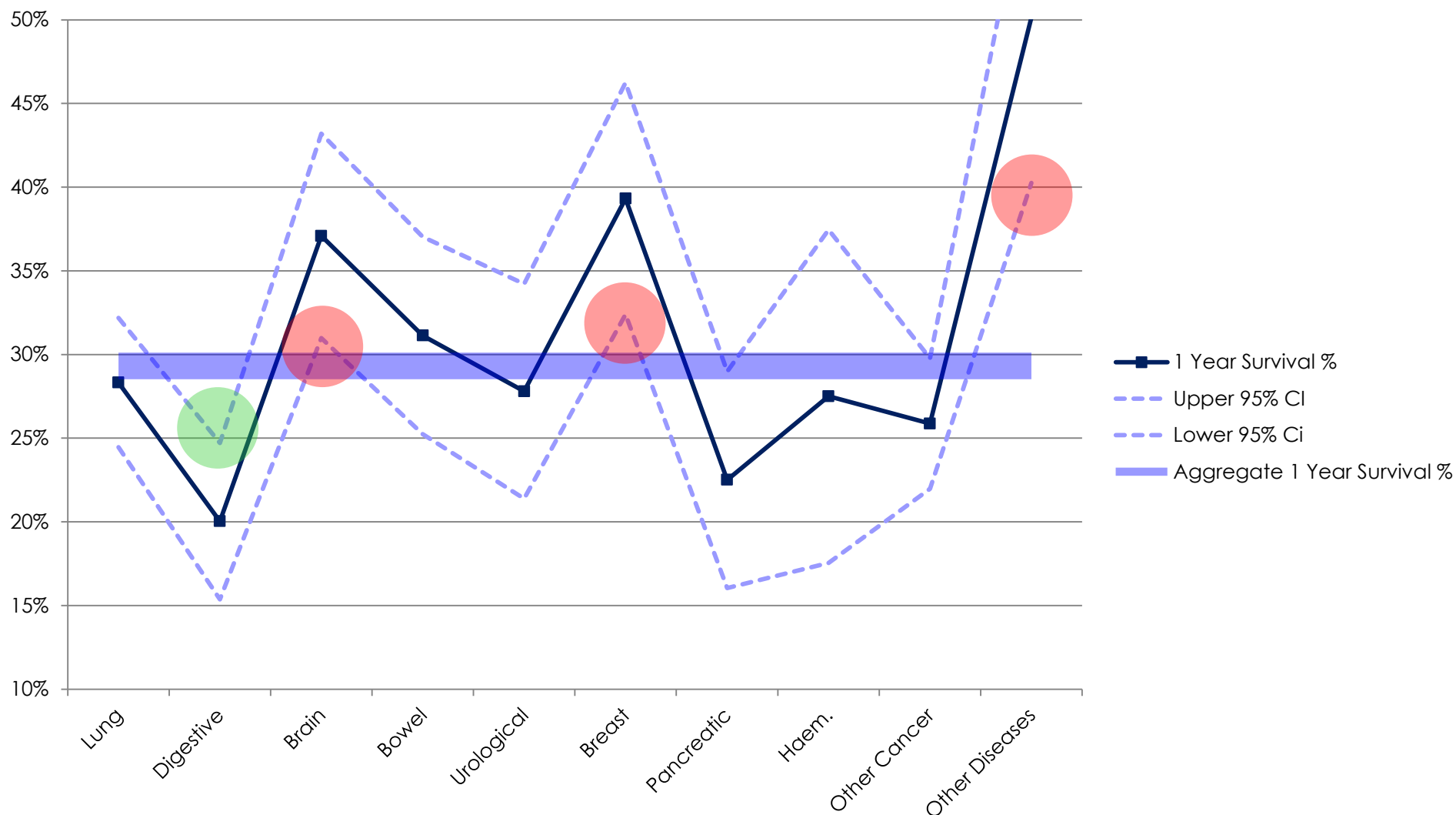
Results

Condition specific survival – 1 Year



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1 Year Survival by Cause of Claim



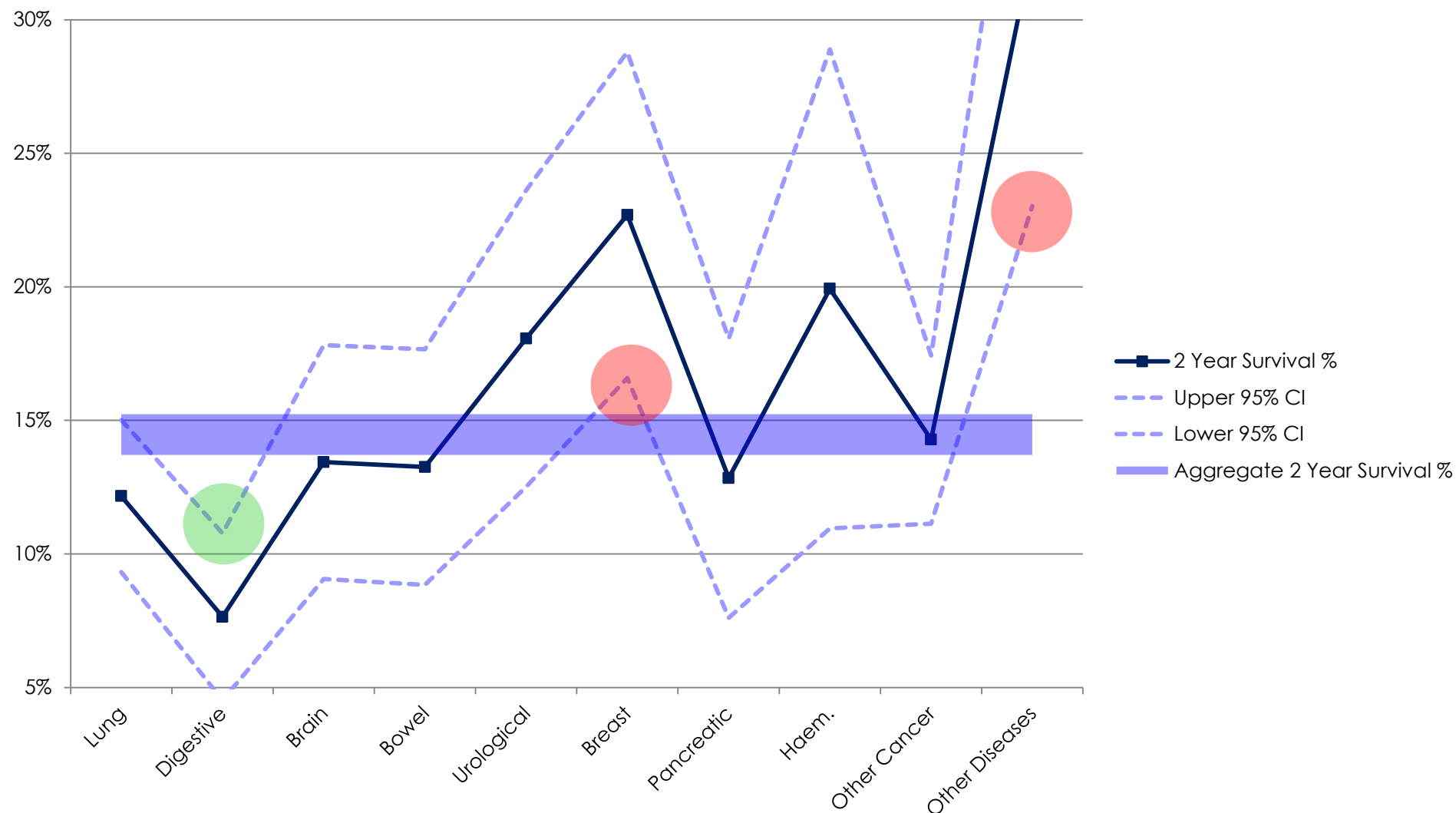
Results

Condition specific survival – 2 Year



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2 Year Survival by Cause of Claim





Questions?